

# Research on walking stairs of wheeled-legged robot based on leg structure transformation

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**Abstract**—When using the quadruped control mode to climb and descend stairs, the common wheeled-legged robot is prone to overturn due to the collision between the knee joint and the stairs. In this paper, virtual model control method is used as the basic control method, and the deformation mode of leg configuration is designed through the wheeled characteristics of wheeled-legged robot. It can select its Elbow or Knee structures for the ascending and descending stairs. All this work obtains satisfactory results the Simulink simulation.

**Index Terms**—Wheeled-legged robot, virtual model control, motion control, structural transformation, up and down stairs.

## I. INTRODUCTION

The existing wheeled-legged robot mainly has two implementation methods of climbing steps: one is to integrate the wheels and legs through its own structural [1]. The robot will expand the wheel and deform it into legs when it needs to cross obstacles. Driving the legs to continuously contact the steps through the rotation of the wheel. The representative robot is shown as the "Land Devil Ray" [2](Fig.1(a)), "FUHAR" [3](Fig.1(b)), "Wheeler" [4](Fig.1(c)) and Engaging drive [5](Fig.1(d)). But they all have a problem that they lack the dexterity of control and are difficult to overcome complex terrain.

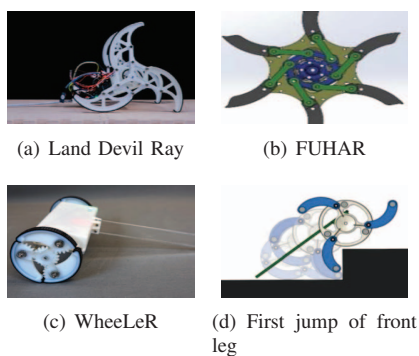


Fig. 1. First kind of wheeled-legged robot

The second is to take the "Swiss-mile" robot [6] as an example(Fig.2(a)). They install wheels on the foot based on the "Anyal" robot. It can walk in the form of wheels and has the flexibility of a foot robot, so it can use the method of walking up steps as a quadruped robot. Similar to the lunar rover, this

robot is suitable for occasions requiring high stability and flexibility, such as the "ATHLETE" [7] hexapod robot(Fig.2(b)), six wheeled biped robot "ASTERISKH" [8](Fig.2(c)), "Transleg" [9](Fig.2(d)), Armadillo [10](Fig.2(e)), hexapod walking robot "Halluc II" [11](Fig.2(f)) and so on.

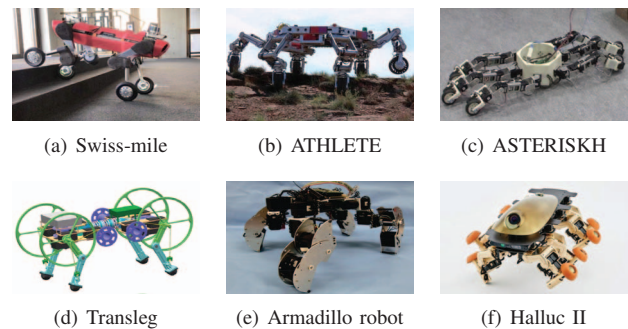


Fig. 2. Second kind of wheeled-legged robot

But this kind of robot has the same problem of up and down stairs as quadruped robot. When the elbow leg goes down the stair, the calf part is easy to collide with the edge of the upper stair. Knee type legs are easy to collide with the edge of the next stair when going up the stair [12] (as the Fig.3).



Fig. 3. Knee bump

Therefore, based on the characteristics of wheeled-legged, this paper proposes a method of climbing up and down stairs of wheeled-legged robot by changing the knee or elbow structure of its legs. Firstly, the single leg motion model of the robot is established and the virtual model control method is used to control the posture of the fuselage. Based on the characteristics

of the wheeled foot robot, the movement functions of jumping and folding are proposed to avoid bumping when going up and down the steps. Finally, the simulation is carried out in Simulink, and a good control effect is obtained.

## II. SYSTEM MODEL

The structure of the wheeled-legged robot described in this paper is shown in the figure. Replace the foot end with a wheel structure, select a motor to drive directly, select a small motor and make a small reducer to minimize the increased moment of inertia of the foot. In addition, the motor layout of the front and rear hip joints adopts the front and rear dislocation arrangement to ensure that the front and rear legs will not interfere in the process of leg folding and unfolding.

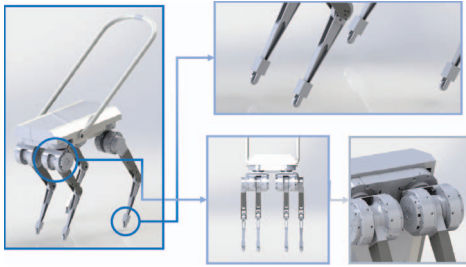


Fig. 4. Robot configuration features

The single leg structure of quadruped robot is simple, and the single leg dynamics can be deduced according to the geometric relationship [13]. The structure of the left front leg is shown in Fig.5, where the coordinate system  $x_0y_0z_0$  is the frame (torso) coordinate system; Coordinate system  $x_1y_1z_1$  is the femoral joint coordinate system,  $x_2y_2z_2$  is the coordinate system of hip joint,  $x_3y_3z_3$  is the coordinate system of knee joint.

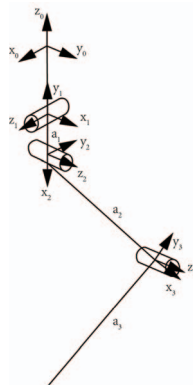


Fig. 5. Single leg coordinate system

Therefore, the single leg kinematics formula of the front leg can be deduced as:

$$\begin{aligned} x &= -(a_2 \sin(\theta_2) + a_3 \sin(\theta_2 + \theta_3)) \\ y &= \sin(\theta_1) (a_1 + a_2 \cos(\theta_2) + a_3 \cos(\theta_2 + \theta_3)) \\ z &= -\cos(\theta_1) (a_1 + a_2 \cos(\theta_2) + a_3 \cos(\theta_2 + \theta_3)) \end{aligned} \quad (1)$$

Through the relationship of forward kinematics, the position of the wheel relative to the trunk can be determined in real time by the rotation angle of the joint motor. The Jacobian matrix  $J$  can be obtained by differentiating the left and right ends of the formula at the same time.

## III. MOTION CONTROL

The behavior sequence of the wheeled-legged robot when facing the stairs is converted into a control flowchart, as shown in the Fig.6. Among them, the main control motion has three parts: leveling - quadruped support control, up stairs - jumping control, down stairs - deformation control.

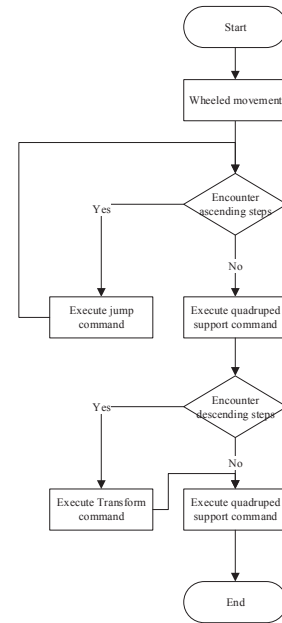


Fig. 6. Control flowchart

### A. Quadruped support stability

The distance between wheel and trunk of a wheeled-legged robot are not rigidly connected during its movement, and will change due to the changes of leg joints. In contrast, you can also control the posture of the trunk by changing the angle of the leg joints during the journey. This requires the robot to have good stability in the four legged support state. In view of the fact that the working environment selected in this paper is not rigorous, the mobility requirements are not high. Therefore, Virtual Model Control(VMC) [14] is selected in this paper.

The main principle is to imagine the force required at the end of the connecting rod to achieve a certain purpose, which is called virtual force. This assumption principle is generally carried out according to the impedance control method, that is  $f_v = K_P (x_e - x_t) + K_I (dx_e - dx_t)$ , where  $K_P, K_I$  is the gain parameter,  $x_e, dx_e$  is the expected displacement and speed,  $x_t, dx_t$  is the actual displacement and speed. The relationship

between joint torque and end output force after virtual force is obtained by

$$\tau = J^T \cdot (-F_V) \quad (2)$$

and solve the torque that a single leg joint should provide.

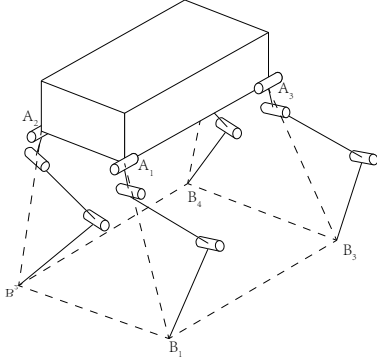


Fig. 7. Schematic diagram of quadruped support

The whole robot under the quadruped support state is as shown in the Fig.7. The leg mass is small and negligible. Therefore, the whole single leg of the robot can be regarded as a two force bar. When the posture of the trunk deviates, it can be considered that a force needs to be applied at the junction of the trunk and the hip joint to drive the posture of the trunk back to the desired state. This force is the virtual force that the single leg hip joint should exert on the trunk. Describe the spatial coordinates of each leg hip as follows:

$$x_i = R * a_i + T \quad (3)$$

which  $x_i$  is the desired hip position of the leg  $i$ .  $R$  is the expected rotation matrix of the torso.  $a_i$  is the spatial coordinate of the hip of the leg  $i$  in the torso coordinate system.  $T$  is the expected displacement vector of the torso.

With the same method, the actual coordinate position of the hip can be obtained. Through the difference between the expected position and the actual position, the virtual force is obtained through PD adjustment, and the joint torque is obtained by substituting into the formula.

### B. Jumping ability

The quadruped controller described in this paper is based on the VMC method, and its essence is to convert the expected foot end displacement into plantar force. If you rashly change the controller to plantar force based on jumping before jumping, it will lead to sudden changes in plantar force and affect the stability of jumping. Therefore, this section considers that the take-off height is changed into the desired displacement of the foot end for control.

When the foot ends leave the ground at the time of the maximum pedal speed, the torso speed is set to  $v_{atthistime_{max}}$ , when the speed is zero, the jumping height reaches the maximum value of  $h$ , and the jumping process is only affected

by gravity, that is, the acceleration is  $-g$ , the expression  $\frac{v_{max}}{2g} = h$ , and the solution is  $v_{max} = \sqrt{2gh}$ ; Let the torso speed accelerate from zero to  $v_{max}$  required the time is  $\frac{T}{2}$ , and the speed expression in the landing stage can be calculated as:

$$v = \frac{1}{2}v_{max} \sin\left(\frac{2\pi}{T}t - \frac{\pi}{2}\right) + \frac{1}{2}v_{max} \quad (4)$$

The speed of the foot end when landing on the ground is also equal to the opposite number of the trunk relative to the ground speed, that is  $v_z = -v$ , the displacement expression of the foot end relative to the torso can be obtained by integrating:

$$z = \frac{1}{2}v_{max} \cos\left(\frac{2\pi}{T}t - \frac{\pi}{2}\right) \frac{T}{2\pi} - \frac{1}{2}v_{max}t \quad (5)$$

When the speed of the foot end relative to the trunk is zero, the displacement of the foot end relative to the trunk reaches the maximum, that is, the leg extension distance reaches the maximum. However, the leg extension distance should be limited to  $z_{max}$ , we can get  $z_{max} = \frac{1}{2}v_{max}t$ , meanwhile  $t = \frac{2z_{max}}{v_{max}}$ .

On this basis, for the hind legs, they do not need to cross together when going up the stairs. So at the end of the landing, the moving distance of the hind legs shall be the moving distance of the trunk relative to the ground when the trunk jumps freely, that is:

$$z_h = -\frac{1}{4}v_{max}T - \frac{(2v_{max} - gt)t}{2} \quad (6)$$

in order to meet the design requirements of climbing the stairs, the front legs should swing to the same height of the stairs to prepare for landing. Select the common stair size  $150mm \times 300mm$ , and the displacement at the beginning of swing is  $-\frac{1}{2}z_{max}$ , speed  $-v_{max}$ , acceleration of 0, displacement of 0.15, speed of 0, acceleration of 0, use quintic polynomial to plan the swing leg trajectory.

### C. Leg configuration transformation

Different from the above support control, the folding and unfolding ability is more inclined to maintain the stability of the body posture and control the additional posture of the single leg to achieve the transformation requirements. Therefore, on the basis of the previous quadruped support control, add the control of wheel steering and single leg joint rotation. Driving the rear legs to move backward through the rotation of wheels, and automatically rotate the rear leg joints under the control of quadruped support until the big and small legs are in the same line; At this time, the posture of the hind leg is the singular posture of a single leg, so it is necessary to add the control of the rotation of the hind leg joint, which is required to rotate its joint rotation angle according to the planned path while maintaining the balance of the posture

of the body in the quadruped support state. First, plan the trajectory of the backward distance as follows:

$$x = \begin{cases} 10 * S * \frac{t^3}{T^3} - 15 * S * \frac{t^4}{T^4} + 6 * S * \frac{t^5}{T^5} & 0 \leq t < T \\ 10 * S * \frac{(2*T-t)^3}{T^3} - 15 * S * \frac{(2*T-t)^4}{T^4} \\ + 6 * S * \frac{(2*T-t)^5}{T^5} & T \leq t \leq 2 * T \end{cases} \quad (7)$$

which  $t$  is the actual time,  $T$  is the time between the backswing and the recovery, and  $S$  is the backswing distance; The joint rotation angle of the hind leg can be obtained by bringing the foot back swing trajectory into the inverse solution of single leg kinematics. After adjustment, it is added to the joint torque obtained in the quadruped support control as the actual torque input of the hind leg joint in the folding change process. Taking the rear leg as an example, the specific folding process example is shown in the Fig.8. Similarly, taking the opposite number can get the forward displacement of the front wheel in the transformation process.

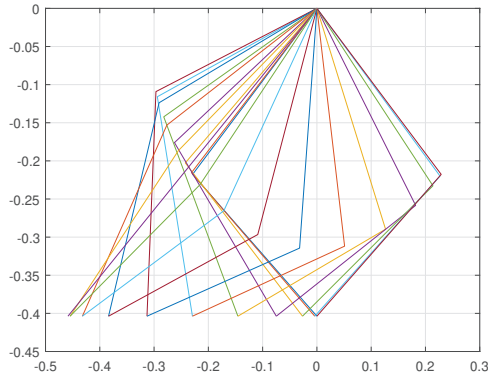


Fig. 8. Hind leg flexion process

The overall control block diagram is as follows(Fig.9).

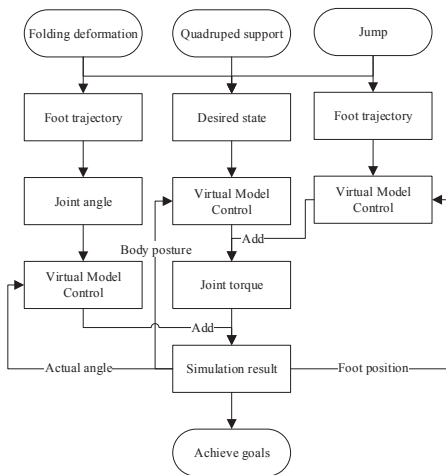


Fig. 9. Simulink logic diagram

#### IV. SIMULATION

The block diagram of the simulation model built in Simulink through the above principles is shown in the Fig.10, including the multi-body dynamics model, support controller, foot move controller and an adjustment module.

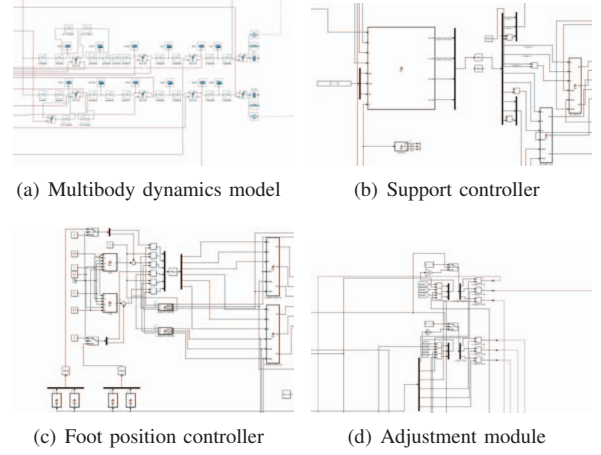


Fig. 10. Simulink model

The simulation effect of jumping in the process of climbing stairs is shown in the Fig.11. Fig.12 is the simulation results.

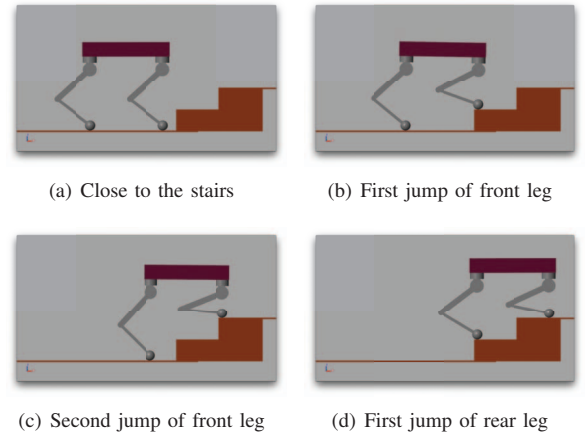


Fig. 11. Jump simulation

The Fig.12(a) shows the six dimensional pose of the robot trunk when climbing the stairs. The pitch rotation and the vertical displacement can quickly return to stability and the lateral displacement, roll rotation and yaw rotation change slightly; Fig.12(b) shows the displacement of the front and rear feet of the robot relative to the trunk when climbing the stairs. Fig.12(c) shows the joint torque of the front and rear legs when jumping up the stairs. It can be seen that the motor whose knee joint is the main output torque in the jumping process has a large instantaneous peak value, which is also the optimization goal of later work; Fig.12(d) shows the contact force between the front and rear legs and the ground during jumping up the stairs. Each take-off will have an obvious landing force

improvement stage, and at the same time, landing will produce a large contact force.

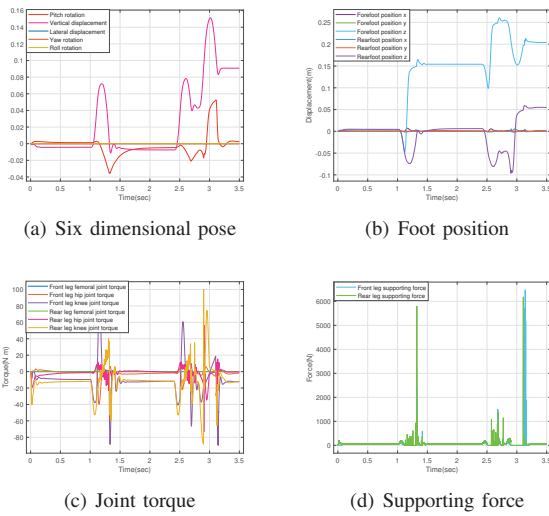


Fig. 12. Jump simulation results

The leg structure transformation after the upper stair is shown in the Fig.13.

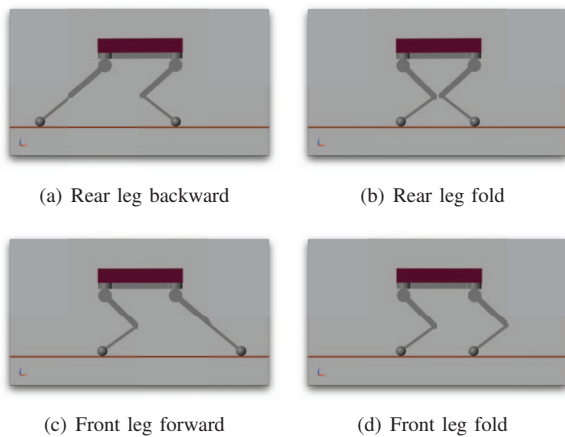


Fig. 13. Transform simulation

Fig.14(a) shows the six dimensional pose of the robot trunk during the folding process. The folding process will not involve the change of posture. It can be seen that the offset in this process is small, which proves the control performance of quadruped support; Fig.14(b) shows the displacement of the front and rear feet driven by the wheel during the folding process, which can better track the smooth and stable trajectory. It can also be seen from the Fig.14(c) that the displacement extension of one leg is the stage of pose fluctuation, and the correction of one leg is also the stage of pose fluctuation returning to stability; The Fig.14(d) shows the joint torque during folding. Except at the beginning of the simulation, the torque change during the folding process is stable without sudden change, and the peak value is small.

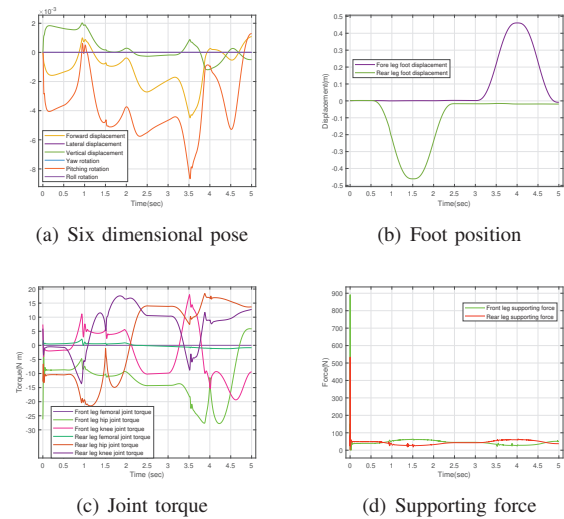


Fig. 14. Transform simulation results

In the process of descending the stair, the stability control of the support is only used as the drive, and the simulation effect is shown in the Fig.15.

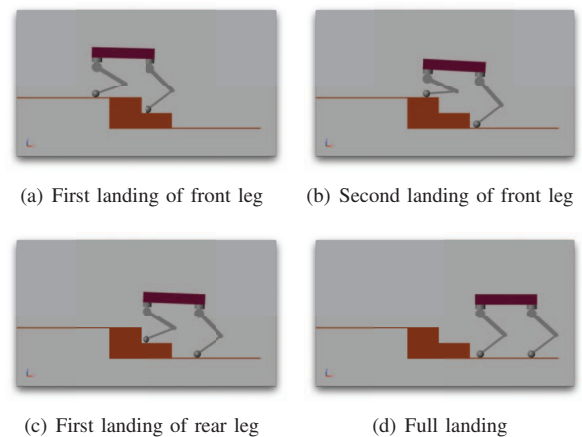


Fig. 15. Simulation of walking down

The Fig.16(a) shows the six dimensional pose change curve of the robot trunk during the process of descending the stairs. In the process of descending the stairs, the pitch rotation angle of the robot will have a certain deviation and the posture will return to the zero after all landing. Fig.16(b) shows the displacement change of the foot end relative to the torso during the process of descending the stairs. The vertical displacement of the front foot decreases and the vertical displacement of the rear foot increases, that is also the process of adjusting the posture to maintain relative balance. Fig.16(c) shows the change of joint torque of front and rear legs during descending stairs. The peak torque can be controlled by reducing the gain value or limiting the output torque; Fig.16(d) shows the contact force curve between the foot end and the stair/ground during the process of descending the stairs. In particular, the front leg



is in the state of downward exploration so resulting in greater impact force.

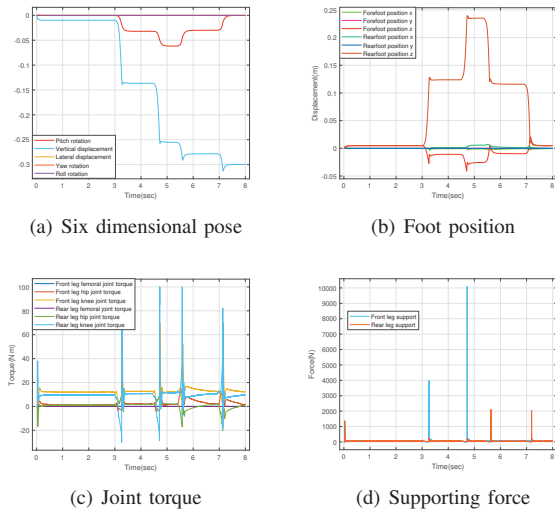


Fig. 16. Simulation results of walking down

The Fig.17 shows the forward distance and speed of the robot in the process of going up and down the stairs. When a single leg of the robot touches each stair, it will produce a certain speed fluctuation and then automatically return to the right. The robot speed is stable at 0.2m/s. The simulation also proves that the wheeled-legged robot can effectively ensure the attitude stability in the process of obstacle climbing by providing forward movement through wheels.

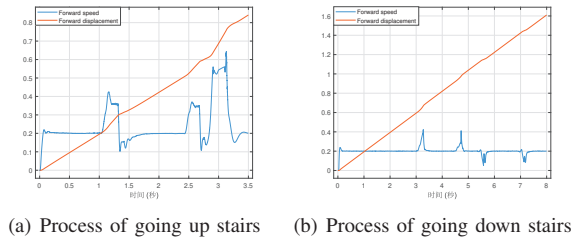


Fig. 17. Forward movement status

## V. CONCLUSION

Based on the characteristics of wheeled-legged robot, this paper proposes a method to improve the performance of walking up and down stairs by changing the knee/elbow structure of legs. This method achieves better control performance through VMC method. On this basis, the next step should be exploring the methods of high speed and improve the leg folding performance. The simulation results prove that compared with quadruped robot, wheeled-legged robot has the following advantages:

- The wheeled-legged robot can keep its body stable in the process of moving the wheels forward, and only needs to jump the legs longitudinally to perform the obstacle

surmounting action. At the same time, discarding the forward jumping force component can further improve the vertical jumping height, and the wheel can also play a buffer role in landing;

- The wheeled-legged robot does not have to raise the foot end when changing the single leg structure, so it can change the leg structure on the premise of maintaining balance;
- Through the transformation of the leg structure, we can choose the appropriate leg's structure to cross the obstacle, which solves the problem that the knee leg or elbow leg is easy to collide with the edge of the obstacle.

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